

COMPUTE ID

WHITEPAPER SERIES

Building ComputeID

Identity infrastructure for AI compute in the age of autonomous systems

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1. Building ComputeID	Identity infrastructure for AI compute - this document
2. Achieving Cryptographic Identity	How to build DevicePassport and AgentPassport at scale
3. Advancing the Protocol	Open standard, decentralisation and community governance
4. Designing for Scale	SDK, API, freemium model and ecosystem growth

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github.com/trustedai/compute-ops/computeid-sdk

Abstract

ComputeID is a cryptographic identity and authentication platform for AI compute infrastructure. In a global GPU rental market projected to reach 162 billion dollars by 2034, any hardware device can claim to be anything - there is zero cryptographic proof of device identity, zero audit trail, and zero accountability. The same crisis is emerging in agentic AI, where autonomous systems operate without verified identity, defined capability boundaries, or immutable audit trails.

ComputeID solves both problems with a unified identity protocol: DevicePassport for GPU and server hardware, AgentPassport for AI agents, and PassportOffice for organisation-wide identity management. Built on NIST 2024 post-quantum cryptographic standards from day one.

Table of Contents

1. Building ComputeID - Introduction	3
2. The Problem in Depth	4
3. Design Requirements	5
4. Core Products	6
5. How ComputeID Works	7
6. Technical Diagrams	8
7. Applications and Impact	13
8. Design Principles	14
9. Addressing Common Concerns	15
10. Roadmap	16

1. The Problem

The AI compute market is experiencing the fastest growth in the history of technology infrastructure. Yet every single one of these devices operates without a verified cryptographic identity. And the AI agents running on them operate with no identity at all.

Any GPU can claim to be any other GPU. Any AI agent can act without accountability. There is no passport system for AI compute infrastructure - and this is the most dangerous gap in the global AI security landscape.

- The GPU identity crisis - hardware fraud, unverified compute, and zero audit trails in a 9.8B market growing to 162B by 2034
- The agentic AI identity crisis - autonomous AI systems acting without verified identity,

defined capability boundaries, or immutable accountability

- The quantum computing horizon - RSA-2048 broken by Shor algorithm within 5-15 years, requiring post-quantum cryptography today

Who built this agent?	No cryptographic mechanism to verify agent creator	Accountability gap - impossible to assign liability
What is it allowed to do?	No standard capability model for AI agents	Scope creep - agents exceed intended function
What has it done?	No audit trail standard for agent actions	Forensic gap - impossible to investigate incidents
How do we stop it?	No standard revocation mechanism	Control gap - no kill switch for rogue agents

2. Core Products

DevicePassport	Cryptographic passports for GPUs and servers. Every hardware device receives a signed certificate - unforgeable digital identity.
AgentPassport	Cryptographic passports for AI agents. Every agent receives an identity, capability certificate, and immutable audit trail.
PassportOffice	Organisation-wide identity management. One registry for all devices and agents, real-time trust monitoring, compliance reporting.

Every GPU needs a passport. Every AI agent needs an identity. ComputeID issues them.

3. Design Requirements

Universal	Any GPU, server, TPU, FPGA or AI agent can receive a certificate regardless of manufacturer
Cryptographically Verifiable	Every identity must be cryptographically provable - not just asserted
Unforgeable	Certificates computationally impossible to forge or transfer
Revocable	Any identity revocable in real time within 60 seconds

Quantum-Safe	All operations use hybrid classical and post-quantum algorithms from day one
Auditable	Every action logged in tamper-evident append-only audit trail
Capability-Bounded	AI agents receive capability certificates defining what they can and cannot do
Open Standard	Protocol specification is open source and freely implementable

4. How ComputelD Works

4.1 Certificate Authority Hierarchy

Root CA	RSA-4096. Self-signed. 10-year validity. Stored offline in HSM. Never used for direct signing.
Intermediate CA	RSA-4096. Signed by Root CA. 5-year validity. Used for all device certificate signing.
Device Certificates	RSA-2048. Signed by Intermediate CA. 1-year validity. Contains quantum-safe extension.

4.2 Quantum-Safe Security

CRYSTALS-Dilithium3 (FIPS 204)	Digital signatures. NIST Level 3. 128-bit post-quantum security.
CRYSTALS-Kyber768 (FIPS 203)	Key encapsulation. NIST Level 3. 128-bit post-quantum security.
Hybrid approach	Classical and post-quantum in every certificate. Backward compatible, forward secure against quantum attacks.

5. Technical Diagrams

Figure 1 - Why compute identity is critical

1. Why identity is critical	Any GPU can claim any identity. Hardware fraud scales with compute. Post-quantum protection built in from day one.
2. The identity gap today	Every GPU rental and every agent deployment happens with zero cryptographic identity or accountability.
3. How ComputeID addresses concerns	Open standard eliminates central control. MPC removes central databases. Three lines of Python to start.

Figure 1. *Why compute identity is critical - the identity gap, zero accountability today, and how ComputeID addresses common concerns.*

Figure 2 - Benefits of ComputeID

Prevent hardware fraud	No provider can misrepresent GPU specs. Every certificate is bound to exactly one device.
Accountable AI agents	Every agent has a defined identity, bounded capabilities, and an immutable audit trail.
Regulatory compliance	EU AI Act Article 12, SOC2 Type II, and NSA CNSA 2.0 - all addressed by one protocol.
Instant revocation	Any device or agent revoked in real time. Propagates within 60 seconds.
Quantum-safe from day one	CRYSTALS-Dilithium3 and Kyber768. Certificates issued today secure against quantum computers.
Open standard	Protocol is open source. Any organisation can implement a ComputeID-compatible CA.

Figure 2. *Key benefits of the ComputeID network across device identity, agent identity, compliance, and security.*

Figure 3 - Common concerns and solutions

Concern: Creates central control	Solution: Open standard. Phase 2 Threshold CA uses MPC - no single point of failure.
Concern: Enables surveillance	Solution: Logs owned by deploying org. Records action types, not content.
Concern: Too complex	Solution: pip install computeid-sdk. Three lines of Python. Cryptography fully abstracted.
Concern: Post-quantum unnecessary	Solution: Certificates last 5-10 years. Build quantum-safe now at zero extra cost.
Concern: Requires central database	Solution: Multi-party computation removes need for any central database.
Concern: Black market for credentials	Solution: Device certs hardware-bound. Agent passports include cryptographic fingerprints.

Figure 3. *Common concerns about compute identity infrastructure and direct answers from the ComputeID protocol.*

Figure 4 - Design requirements across three stages

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Universal eligibility	Multi-party uniqueness check	JWT token 1-hour expiry
Cryptographic verifiability	Secure compute	Replay attack prevention
Unforgeable binding	No centralised database	Capability verification
No upload of sensitive data	OCSP real-time revocation	Immutable audit logging
Multi-issuer support	Recovery mechanism	Instant kill switch

Figure 4. *Design requirements across the three stages of the ComputelD protocol.*

Figure 5 - DevicePassport registration and authentication flow

A simplified diagram of how a device joins the ComputelD network. After registering, the CA issues a signed certificate. An admin approves and a JWT token is issued valid for one hour.

Figure 5. *DevicePassport flow. Device registers, CA issues signed certificate, admin approves, JWT token issued for 1 hour. Every subsequent connection authenticated and logged.*

Figure 6 - AgentPassport issuance and lifecycle

The AgentPassport lifecycle from issuance to capability definition, action logging, and instant revocation.

Figure 6. *AgentPassport lifecycle. Issue passport, set immutable capabilities, log every action automatically, revoke instantly with `passport.revoke()` if behaviour is unexpected.*

Figure 7 - Identity approach comparison

Comparison of identity approaches evaluated against ComputelD design requirements. Green indicates meets requirement, red indicates insufficient.

Figure 7. *Identity approach comparison. Only ComputelD satisfies all five design requirements including cryptographic proof, agent identity, audit trail, quantum safety, and fast revocation.*

Figure 8 - ComputelD network architecture

Devices and agents on the left register with the ComputelD CA. The CA issues quantum-safe certificates and maintains the PassportOffice registry. Relying parties on the right verify certificates and access audit reports.

Figure 8. *ComputeID network architecture. Three zones: Devices and Agents, ComputeID Network with CA and PassportOffice, and Relying Parties including GPU rental companies, enterprise clients, AI platforms and regulators.*

6. Regulatory Alignment

EU AI Act (2026)	Article 12 requires detailed logging for high-risk AI systems. AgentPassport provides immutable action logs for every agent action.
NSA CNSA 2.0 (2030)	Mandates post-quantum cryptography. ComputeID is already CNSA 2.0 compliant with Dilithium3 and Kyber768.
NIST AI RMF	Requires provenance and accountability. ComputeID provides full provenance chains for all devices and agents.
SOC2 Type II	Requires logical access controls and audit logging. DevicePassport provides authenticated device access and audit trails.

7. Design Principles

1. Zero Trust	No device or agent is trusted by default. Every identity must be cryptographically verified on every connection.
2. Least Privilege	Every AI agent receives only the minimum capabilities required. Capability boundaries are immutable once issued.
3. Immutable Audit	Every action is logged with a cryptographic commitment. Audit logs are append-only and tamper-evident.
4. Instant Revocation	Any identity revocable in real time. Revocation propagates within 60 seconds. No grace period.
5. Quantum Safety	All cryptographic operations use hybrid classical and post-quantum algorithms. Post-quantum is a day-one requirement.

6. Open Standard	The ComputeID protocol specification is open source. Any organisation can implement a compatible CA.
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8. Roadmap

Phase 1 - Foundation (Complete)	DevicePassport, AgentPassport, PassportOffice, hybrid PQC certificates, open source SDK v1.1.0, REST API, managed CA
Phase 2 - Enterprise (Q3 2026)	Threshold CA with MPC, CLI tool, MCP server integration, Stripe billing, free tier portal, GitHub Actions
Phase 3 - Standards (Q1 2027)	IETF RFC submission, IEEE working group, full liboqs PQC certificates, multi-tenant platform, compliance dashboard
Phase 4 - Decentralisation (2028)	Community-governed CA infrastructure, distributed node operators, on-chain certificate transparency

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github.com/trustedaicompute-ops/computeid-sdk